

Real Estate CRM: Comprehensive API & Deployment Documentation

1. API Documentation

Base URL: <http://localhost:4000/api>

Authentication & Users

- **POST /auth/register**: Create a new account.
- **POST /auth/login**: Obtain a JWT token.
- **GET /users**: (Admin) List all system users.
- **PATCH /users/:id/role**: (Admin) Update user permissions.

Acquisitions (Leads)

- **GET /leads**: Retrieve leads (Admins see all; Agents see assigned).
- **POST /leads**: Register a new prospect.
- **POST /leads/webhook**: Public endpoint for web forms.
- **PATCH /leads/:id/status**: Transition lead status.
- **PATCH /leads/:id/assign**: (Admin/Manager) Assign lead to agent.

Property Portfolio

- **GET /properties**: List curated property listings.
- **POST /properties**: Add a new asset to the portfolio.
- **POST /properties/:id/images**: Upload asset imagery.

Client Management

- **GET /clients**: List all buyer/seller profiles.
- **POST /clients**: Create a new client profile.
- **GET /clients/:id**: Retrieve specific client details.
- **PUT /clients/:id**: Update client information.

Pipelines (Deals)

- **GET /deals:** List active transaction pipelines.
- **POST /deals:** Initiate a new transaction.
- **PATCH /deals/:id/stage:** Move deal through Kanban stages (Negotiation, Agreement, Closed).
- **GET /deals/:id/commission:** Calculate agent commission for the deal.

Activities & Follow-ups

- **GET /activities:** Retrieve interaction history log.
- **POST /activities:** Log a new interaction (Call, Meeting, Email).
- **POST /followups:** Schedule a future follow-up task.
- **PATCH /followups/:id/done:** Mark a follow-up task as completed.

Intelligence & Analytics

- **GET /reports/overview:** High-level KPI summary.
- **GET /reports/monthly:** Monthly revenue and deal volume trajectories.
- **GET /reports/agents:** (Admin) Comparative agent performance metrics.
- **GET /reports/export:** Download comprehensive Excel performance dossier.

2. Admin vs Agent Visibility

The system utilizes granular role-based access control (RBAC) to manage data visibility:

Feature	Admin / Manager View	Agent View
Acquisitions	Full visibility of all system leads.	Visibility limited to assigned leads.
Pipelines	Oversight of all active deals.	Management of personal deals only.
Intelligence	Global fiscal and performance reports.	Personal commission and unit targets.
Users	Full user and role management.	No access to user management.

3. Deployment & CI/CD

Infrastructure Orchestration

- 1. Environment Setup:** Configure *server/.env* with DATABASE_URL, JWT_SECRET, and SMTP/Twilio credentials.
- 2. Launch:** Run docker compose up -d --build to instantiate the stack.
- 3. Routing:** Nginx handles production React routing via *try_files* redirection to index.html.

Security & Persistence

- **Data Persistence:** PostgreSQL data is stored in the *pgdata* Docker volume.
- **Encryption:** Passwords are salted and hashed via BCrypt. JWT tokens secure sessions.
- **CI/CD:** A GitHub Actions pipeline is configured for automated testing and build verification.